

Complex Variable Wing Designs and Aerodynamic Performance

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Abstract—This paper presents a comparative study of several variable and non-conventional aircraft wing designs, focusing on their aerodynamic performance. The analysis emphasizes the underlying principles of lift generation, angle of attack effects, and stall behavior, providing a theoretical foundation for understanding wing efficiency. Five wing configurations—swing (variable-sweep) wings, forward-swept wings, delta wings, and swept wings—are reviewed in terms of lift-to-drag ratio, stability, and adaptability across flight regimes. The study synthesizes findings from prior experimental and computational research to highlight the advantages and limitations of each wing type. Literature results indicate that variable (morphing) wings offer the greatest flexibility and efficiency for multiple flight conditions, while traditional delta and forward-swept designs provide specialized performance benefits in high-speed regimes. Additionally, a computational case study of the ONERA M6 swept wing is included to show the transonic swept-wing flow physics and shock-induced aerodynamic effects. This report serves as a foundational reference for the design and optimization of advanced aerodynamic structures in both manned and unmanned aircraft applications.

DEFINITIONS

Re	Reynolds number (dimensionless)
M	Mach number (flight speed / speed of sound)
α	Angle of attack (degrees)
C_L	Lift coefficient (dimensionless)
C_D	Drag coefficient (dimensionless)
L/D	Lift-to-drag ratio (aerodynamic efficiency)

I. INTRODUCTION

All of the aircraft wings analyzed in this paper follow the same fundamental principles of flight. Primarily, an airplane generates lift by creating a pressure difference between its upper and lower surfaces. Aircraft take advantage of Bernoulli's principle, which states that faster-moving air has a lower pressure. A wing is typically shaped so that airflow travels faster over the upper surface than over the lower surface. The wing's shape constricts ("squeeze") the airflow and a low pressure zone pulls air at higher speeds. The wing's angle of attack (α) -the angle between the wing and the incoming airflow—also impacts lift through Newton's Third Law where air "pushes" the wing upwards.

Pilots must also be mindful of the angle of attack (α) because if it becomes too steep, depending on the wing design, the airflow will separate from the surface. This separation disrupts the low-pressure region on the upper surface, causing lift to decrease rapidly and potentially resulting in an aerodynamic stall. Increasing the angle of attack increases lift only

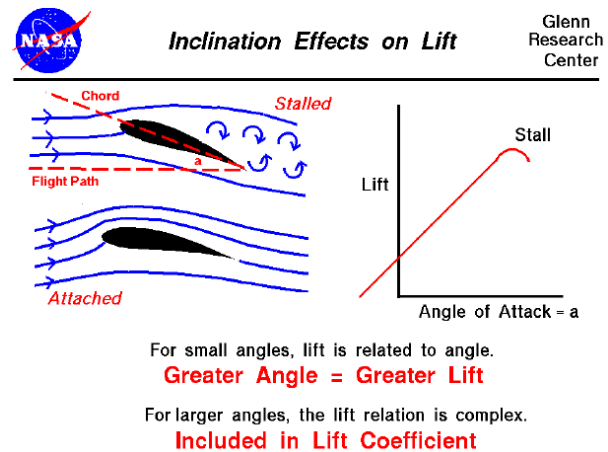


Fig. 1. Effect of angle of attack on lift coefficient and stall behavior.

up to a critical point, beyond which the wing stalls. Figure 1 illustrates this behavior and shows the relationship between angle of attack and the lift coefficient (C_L).

II. MATERIALS AND METHODS

This research paper compares and contrasts 5 different types of wings, evaluating the aerodynamic performance under different conditions. This study relies almost entirely on technical literature rather than experimentally derived information. Primary sources include NASA literature and aircraft design textbooks. We analyzed values such as the Reynolds Number, Mach number, and coefficient of lift, all in an attempt to highlight the use case of all of these wings.

III. RESULTS AND DISCUSSION

In this section, we present and compare the aerodynamic performance characteristics of five variable or non-conventional wing designs.

A. Swing (Variable-Sweep) Wings

The Grumman F-14 Tomcat was a highly advanced carrier-based interceptor designed to meet demanding and often conflicting performance requirements. Its variable-sweep wing system allowed the aircraft to adapt its aerodynamic configuration in flight, enabling both low-speed stability for aircraft carrier operations and high-speed efficiency for long-range interception missions. This design made the F-14 a standout

example of how aerodynamic theory and mechanical engineering can be combined to solve real-world aerospace challenges.

At high speeds, the F-14 sweeps its wings rearward to reduce wave drag and improve performance in the transonic and supersonic regimes. Sweeping the wings decreases the component of airflow normal to the leading edge, effectively increasing the critical Mach number and delaying shock wave formation. This capability allowed the aircraft to sustain efficient high-Mach flight while maintaining stability and control.



Fig. 2. F-14 Tomcat in high-speed flight with wings swept back

The wing-sweep system relied on a central gearbox that mechanically synchronized both wings, ensuring symmetrical motion under high aerodynamic loads. Each wing pivoted within a reinforced wing glove structure designed to withstand significant bending and torsional stresses. This mechanical synchronization was critical for flight safety, as asymmetric wing positions could cause severe roll instability, especially at high speeds.

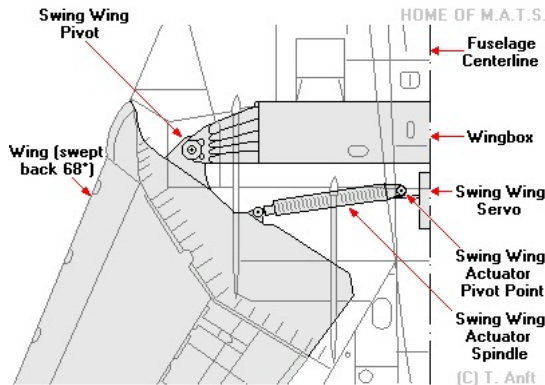


Fig. 3. Central wing-sweep gearbox and pivot mechanism

From a fluid mechanics perspective, the F-14 variable-sweep wing provides a clear demonstration of compressible flow effects. By altering wing sweep, the aircraft reduced the effective Mach number experienced by the wing, delaying shock formation and reducing drag associated with compressibility. The design also highlights key concepts such as lift-drag tradeoffs, boundary-layer behavior, and aeroelastic effects, making the F-14 an effective real-world case study for applying fluid mechanics principles in aerospace engineering.

B. Forward-Swept Wings

The forward-swept wing, also called the reverse-swept wing, is a plane design where the quarter chord line of the

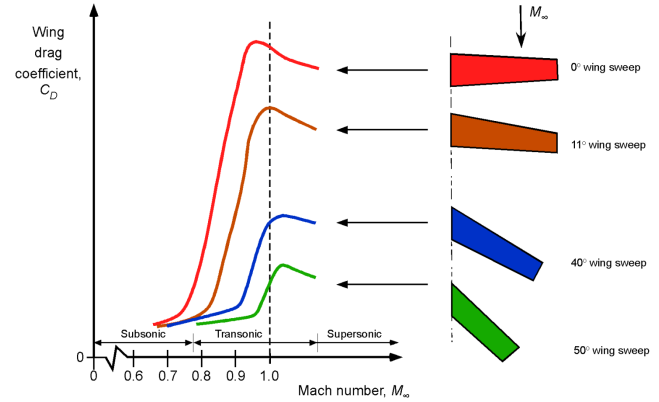


Fig. 4. Effect of wing sweep on airflow direction and effective Mach number

plane has a forward sweep. They are designed for higher maneuverability at higher angles of attack, but are also more unstable.

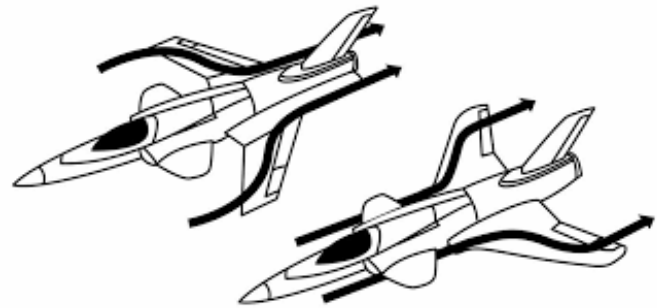


Fig. 5. Airflow in forward-swept wings (left) vs. swept-back wings (right).

1) *Inward Airflow*: In conventional swept-back wings, pressure is higher near the body and lower near the wing tip, which pushes the boundary layer air outwards. Friction slows air near the surface, causing the boundary layer air to lose energy. If there is too much energy loss, this leads to flow separation where the air can no longer follow the shape of the wing and peels away, which causes drag, reduces lift, and can reduce the effectiveness of control surfaces like ailerons that depend on a smooth airflow. In contrast, forward-swept wings direct spanwise airflow inwards towards the fuselage. Because of this, the wing root will stall first, while the wing tips and control surfaces remain in relatively smooth airflow. This allows for better roll control, making forward-swept wings highly maneuverable at high angles of attack. Research shows that compared with swept-back wings, forward-sweep wings have a higher lift coefficient at angles of attack greater than 26° . The lift curve also changes more gradually compared to the sharp drop in lift for swept-back wings, displaying improved stall performance.

2) *Aeroelastic Divergence*: The primary challenge for forward-swept wings is aeroelastic divergence. For longitudinal stability, the center of gravity typically should be in front of the aerodynamic center, creating a nose-down pitching tendency that the tail counteracts. However, for forward-swept designs such as the Grumman X-29, an American experimental

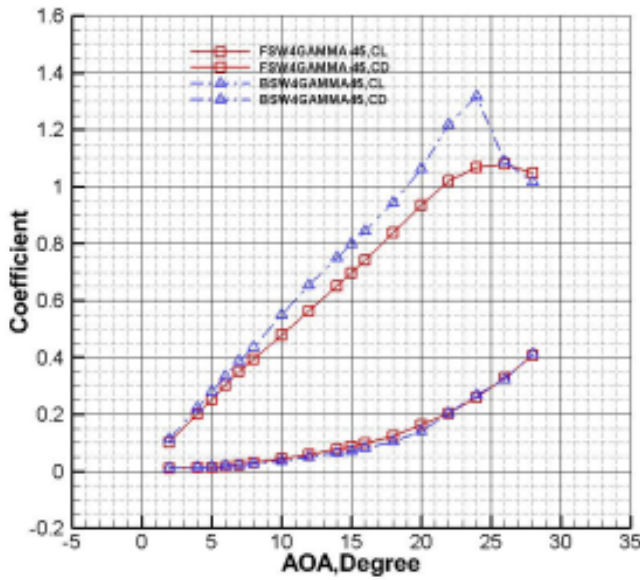


Fig. 6. Lift coefficient of forward-swept wings (red) and swept-back wings, from AOA of 2 to 28 degrees at Mach = 0.2 and Re = 4.6 million.

aircraft, the center of gravity is aft (behind) the aerodynamic center, creating aerodynamic moments that force the nose to pitch up. This increases the angle of attack of the plane. Lift increases with angle of attack, which further pitches the nose up, resulting in an undesirable runaway positive feedback loop that can continue until the wing passes its elastic limit and structurally fails. Going from subsonic to supersonic flight resulted in a 35% negative static margin (pitch instability) for the X-29.

To better control the aeroelastic twisting while keeping it lightweight, the X-29 incorporated composite materials like carbon fiber into its wings. Additionally, it is controlled by an extensive fly-by-wire avionics control system with six digital computers and three analog backups to sense and adjust control surfaces up with up to 40 commands per second. Despite this, the risk of system failure in the X-29 was equivalent to the risk of mechanical failure in a conventional system, and forward-swept wing designs remain rare due to engineering challenges.

C. Delta Wings

1) *Delta Wing Characteristics*: Delta wings are wings shaped in a triangular (just like the greek letter Δ) shape. The wing's leading edge features a large sweep angle. Its chord is taken to be the average of root-tip distances across the wing named the "Mean Aerodynamic Chord (MAC)". Most of the wing's area is close to the fuselage making it possible to build a stronger and stiffer wing while making it even lighter than an equivalent wing. This makes it easier and even more economically feasible to build [1].

As described in the Introduction, airplane wings, including but not limited to delta wings, use Bernoulli's principle to create a pressure difference between its upper and lower



Fig. 7. Dassault Rafale

surface and therefore generate lift. However, delta wings take advantage of the concept of vortices.

A delta-winged aircraft generates vortices to employ vortex-induced lift at high angles of attack (α). The main idea of the delta wings is to control these vortices and take advantage of them to generate lift. Certain additions to the wing can be implemented to do this such as a retractable "moustache" or fixed leading-edge root extension (LERX) [2]. "Vortex development depends on leading-edge sweep (ϕ) and angle of attack (α)" [3].

A sharp leading edge on a delta wing helps separate airflow and creates more stable leading edge vortices which are used to generate lift. Sharp leading edges make it easier to predict and control where the vortices will form to improve aircraft lift. However, some aircrafts like the Concorde and other early fighters however use a curved edge to perform better at reduced velocities.

The trade-off from relying on vortices lies on the additional drag experienced. Nonetheless the effect of drag is countered by the advantages of delta wings such as increased maneuverability, a more stable flight at low speeds with high angles of attack, and reduced drag and increased control at supersonic speeds [1].

2) *Advantages of Delta Wings*: The relationship between lift (C_L) and angle of attack (α) for a delta wing is characterized by their high maximum (C_L) values at large (α) values and can reach higher angles of attack before stalling. The slope of this curve is more gentle with lower lift coefficients for low angles of attack, but the wing thrives at high AoAs. Figure 8 shows this C_L vs. α curve through experimental values [3].

Delta wings also thrive at high speeds, especially during transonic (close to speed of sound) and supersonic flight due to its large sweep angle (angle between line perpendicular to fuselage and wing leading edge). This large sweep angle delays the supersonic shockwaves formed as the airplane reaches Mach 1. It experiences a slower increase in drag as it approaches the speed of sound to avoid the sudden "drag divergence" which causes max drag during transition to supersonic speeds as shown in Figure 9.

H. Rahman et al. (2013) published a paper on ANSYS computational fluid dynamics done on delta wings using ANSYS Fluent-14. Figure 10 as well data and specifications can be found in their paper [3].

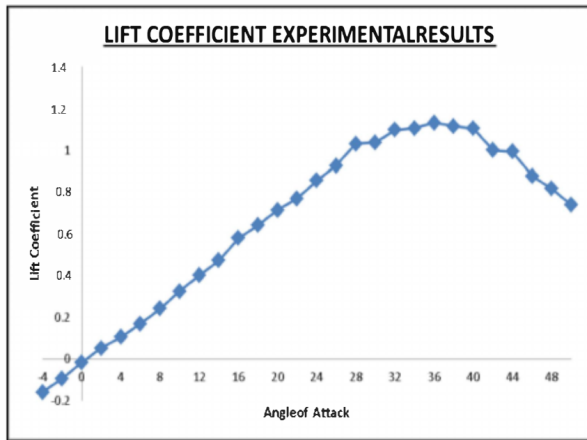


Fig. 8. Experimental Lift Coefficient against Angle of Attack [3].

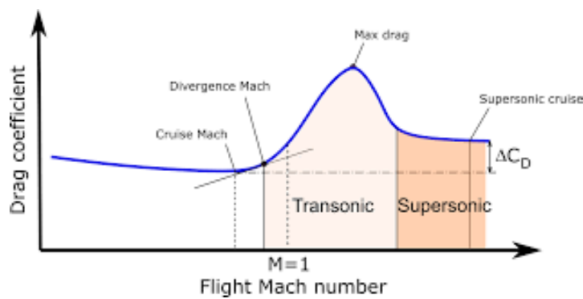


Fig. 9. Drag Coefficient vs. Flight speed. Drag coefficient transitions from subsonic to supersonic speeds with maximum drag at transonic speeds.

3) *Dassault Rafale*: Figure 7 shows the Dassault Rafale, a great example of the Delta Wings in action. This aircraft is a modern, twin-engine fighter jet that uses a "canard delta wing design. (...) The aircraft is capable of withstanding from -3.6 g to 9 g (10.5 g on Rafale solo display and a maximum of 11g can be reached in case of emergency)" [2]. Dassault used a delta wing design as well as a canard (small forewing in front of main wings) to maximize maneuverability.

Its military design allows the aircraft to take off and land in various ways such as catapult launches or aircraft carrier landings. This allows for short launches to reach the necessary speed for a stable flight. The aircraft's design is intentionally aerodynamically unstable since this increases maneuverability, a sought-after feature in military applications. A less stable airplane requires less control inputs for path changes. The Dassault Rafale uses digital fly-by-wire (FBW) controls such that the computer system determines how to move each actuator or mechanical system to maintain the desired path.

4) *Comparison to Regular Wings*: Regular commercial aircrafts do not rely on leading edge vortices to induce lift because it makes the aircraft experience significant drag which in turn increases fuel consumption. Delta aircrafts on the other hand usually undergo supersonic flight and are made to execute maneuvers even if inefficient in terms of fuel consumption.

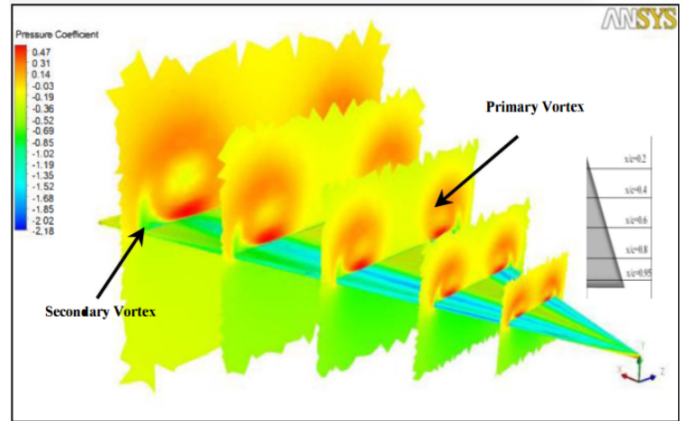


Fig. 10. ANSYS Fluent-14 simulation of delta wings with vortices at visible cross sections within the wing [3].

D. Swept Wings

A swept wing is any wing that is not perpendicular to the fuselage of the plane. They exist in many forms, from delta wings, to forward swept wings. Although early planes almost all had non-swept wings, every modern airliner and fighter jet has a distinctive sweep in their wings. In this section, we will cover the advantages and disadvantages of swept wings by examining two very different aircraft. This first being the most common passenger airliner in the world, the Boeing 737, and the second being the most common plane in the world, the Cessna 172.



Fig. 11. Boeing 737-800



Fig. 12. Cessna 172P

1) *Transonic Drag and Critical Mach*: As an airplane approaches the speed of sound, it encounters a massive amount of drag that far exceeds parasite or induced drag. This force is Wave drag, and it is caused by the fast moving air over the wing. Although a plane may have a mach number of .8 [7], air moving over its wings can move above the speed of sound, creating shock waves. This leads to flow separation, and massive amounts of drag. This drag continues until the air below the wing is also moving at the speed of sound and they create a bow and oblique shock waves.

Critical mach number is the speed at which any portion of the airflow moving over the wing is at the speed of sound. For the Boeing 737, the critical mach number is .78 which is actually slower than its max cruising speed. This is because the wing is designed to handle very small shockwaves without stalling due to its airfoil shape and material. However, there is only so much that can be done structurally, so maximizing critical mach number is a key constraint in designing subsonic aircraft. A non swept wing like that of the Cessna has a much lower critical mach, but it is irrelevant as a Cessna flies below

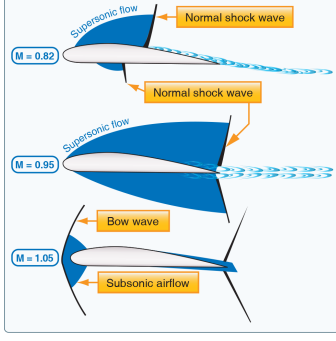


Fig. 13. Transonic flow separation leading to shockwaves and wave drag

.2 mach. The reason the 737 is able to increase its critical mach number is because of the “cosine rule.” This simply is the relationship between the angle of the sweep (Λ) and the cord-wise velocity (V_n). For any sweep V_n is going to decrease meaning the effective mach number at the wing decreases. [8]

$$V_n = V_\infty \cos(\Lambda)$$

2) *Lift Coefficient and Induced Drag*: Despite offering huge upside in critical mach number, swept wings would hinder a plane like the Cessna. If we could take two identical wings but one is swept at an angle (Λ), their coefficients of lift at some given angle of attack would have the following relationship.

$$C_{l,\text{swept}} \approx C_{l,\text{unswept}} \cos(\Lambda)$$

The sweep of a boeing 737-800 is 25 degrees [8] meaning there is a 10% reduction in lift compared to an unswept wing. Of course the 737 is flying so much faster that it is able to produce the necessary lift, but it is not realistic to land at mach .8. It is for this reason that larger planes require aerodynamic devices like flaps. These aid the plane in producing more lift for takeoff and landing, allowing the plane to liftoff and touchdown in safe distances. The Cessna has no need for these because it is a fraction of the weight going a fraction of the speed.

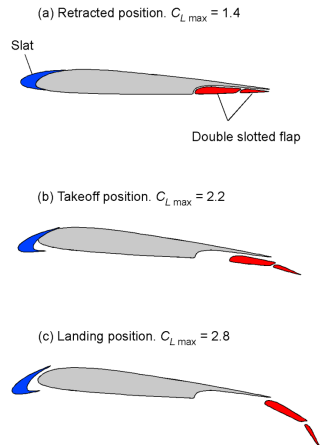


Fig. 14. Aerodynamic devices altering the airfoils profile leading to an increased coefficient of lift

Induced drag is caused by airfoil moving through the air and creating lift. It is inversely proportional to the aspect ratio (AR) or the ratio of a wing's length to the average cord. The cessna is moving so slowly that its AR is not very relevant as it is more important to optimize for span efficiency at that scale. The Boeing's swept wing reduces AR but ultimately pays dividends at higher speed regimes. [9]

3) *Boundary Layer and Stall Characteristics*: As mentioned previously, when a wing is swept, its critical mach number decreases because it is a function of the cord-wise velocity. Another component is also derived and it is orthogonal V_n . This component is the spanwise velocity (V_s) and it is zero for an unswept wing

$$V_s = V_\infty \sin(\Lambda)$$

On a 737, this component moves low energy boundary layer air towards the ends of the wings. This causes a rapid buildup at the wingtips leading to flow separation. A Cessna on the other hand, will begin its stall at the root of the wing first leaving stable airflow over the control surfaces at the ends of the wings. This allows the pilot to remain in control of the aircraft. The 737 on the other hand, is more of a passenger when a stall occurs. Additionally due to the sweep, stalling in a 737 effectively moves your center of lift forwards, pitching the plane up and worsening the situation. Of course, 737s do not just fall out of the sky randomly and that is due to many safety features such as vortex generators and wing fences. [10]

E. Computational Case Study: ONERA M6 Swept Wing

a) *Motivation*: The following CFD simulation of the ONERA M6 wing was done as a controlled swept wing configuration. The ONERA M6 wing is not actually flown on any aircraft but was created for studies of transonic swept-wing flow phenomena and has since been used as a standard configuration for experiments and computations to validate results. All of the geometry data and other information are public, so direct comparison with published data can be made. Therefore, the ONERA M6 wing configuration is an ideal candidate for use as a standard to validate the methods used to draw conclusions about actual wings from aircraft.

b) *Flow Conditions and Freestream Properties*: The simulation corresponds to ONERA Case 2308 with freestream Mach number $M_\infty = 0.84$ and freestream static temperature $T_\infty = 300$ K. For air, $\gamma = 1.4$ and $R = 287$ J/(kg · K). The freestream speed follows from

$$M = \frac{u_\infty}{a_\infty}, \quad a_\infty = \sqrt{\gamma R T_\infty}, \quad (1)$$

so

$$u_\infty = M_\infty \sqrt{\gamma R T_\infty} = 0.84 \sqrt{1.4 \cdot 287 \cdot 300} \approx 291.6 \text{ m/s}. \quad (2)$$

In this case, the Reynolds number is stated to be $Re = 11.72 \times 10^6$, based on the mean aerodynamic chord

$$L = 646.07 \text{ mm} = 0.64607 \text{ m}. \quad (3)$$

The dynamic viscosity at the freestream temperature was computed using Sutherland's law,

$$\mu(T) = \mu_{\text{ref}} \left(\frac{T}{T_{\text{ref}}} \right)^{3/2} \left(\frac{T_{\text{ref}} + S}{T + S} \right), \quad (4)$$

with typical air constants $\mu_{\text{ref}} = 1.716 \times 10^{-5} \text{ kg}/(\text{m} \cdot \text{s})$, $T_{\text{ref}} = 273.15 \text{ K}$, and $S = 110.4 \text{ K}$. Evaluating at $T = 300 \text{ K}$ gives

$$\mu \approx 1.85 \times 10^{-5} \text{ kg}/(\text{m} \cdot \text{s}). \quad (5)$$

Using the Reynolds number definition

$$Re = \frac{\rho u_{\infty} L}{\mu}, \quad (6)$$

the freestream density was calculated to be

$$\rho_{\infty} = \frac{Re \mu}{u_{\infty} L} \approx \frac{(11.72 \times 10^6)(1.85 \times 10^{-5})}{(291.6)(0.64607)} \approx 1.15 \text{ kg}/\text{m}^3. \quad (7)$$

Finally, freestream static pressure was calculated using the ideal gas law,

$$p_{\infty} = \rho_{\infty} R T_{\infty} \approx (1.15)(287)(300) \approx 9.89 \times 10^4 \text{ Pa}. \quad (8)$$

These freestream properties were used consistently in the pressure far-field boundary condition and in the solver reference quantities.

c) Turbulence Model and Boundary Conditions: The flow was modeled as steady, compressible, and turbulent flow using a pressure-based solver. Air was modeled as an ideal gas with temperature-dependent viscosity. Turbulence closure was performed using the one-equation Spalart–Allmaras model. The turbulent viscosity at the far-field boundary was taken to be three times the freestream kinematic viscosity, per standard practice for the case at hand.

A pressure far-field boundary condition was applied at the outer domain boundary. Freestream Mach number, temperature, and flow direction were prescribed directly. The angle of attack of $\alpha = 3.06^\circ$ was imposed by resolving the freestream velocity vector into Cartesian components. The wing surface was modeled as a stationary, no-slip wall. A symmetry boundary condition was used at the wing centerline to reduce computational cost while maintaining three-dimensional flow behavior.

d) Geometry and Computational Domain: First, the ONERA M6 wing geometry was imported from the NASA Langley Turbulence Modeling Resource. The wing was modeled with a sharp trailing edge, as is common in CFD-focused representations of the wing. A hemispherical far-field domain was constructed around the wing and extended several chord lengths in each direction in order to limit boundary effects. A symmetry plane was applied at the centerline so that only half of the flow needed to be simulated.

e) Mesh Generation: An unstructured tetrahedral mesh was generated for the whole domain. The refinement of the mesh was focused near the wing surface and in the near wake using a body-of-influence region to capture large pressure gradients and shock structures expected at transonic conditions. Extra edge and face sizing were used along the airfoil with the purpose of keeping geometric accuracy. Inflation layers were

provided at the wall, with a target wall spacing corresponding to approximately $y^+ \approx 30$. The total element count was kept below one million due to the limitations of the ANSYS Student solver. Resolution was prioritized in regions most relevant to flow physics.

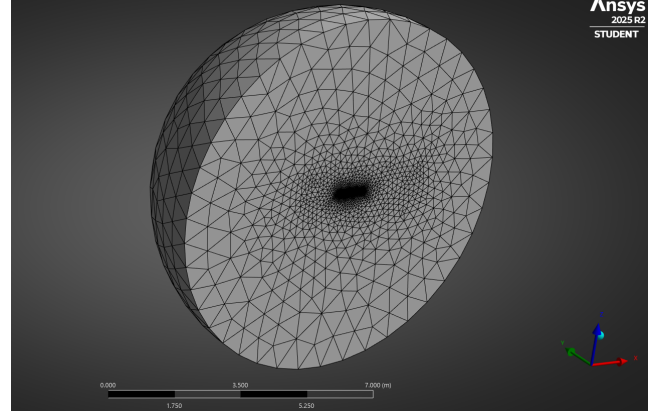


Fig. 15. Unstructured tetrahedral mesh exhibiting near-field refinement around the ONERA M6 wing and progressively coarser resolution toward the far-field boundary.

f) Force Coefficient Definitions: The surface forces were projected onto the directions parallel and perpendicular to the freestream velocity vector, allowing for the calculation of lift and drag coefficient values. For the given angle of attack value, the drag direction was set to be parallel to the incoming flow and the lift direction to be perpendicular. Reference area and characteristic length values were obtained directly from the ONERA M6 validation documentation, allowing consistency with published experimental and numerical results.

g) Results: The convergence of the simulation was achieved within about 200 iterations, but 500 were performed to provide a buffer. The coefficients that are reported by Fluent for the final converged solution are

$$C_L = 0.2539, \quad C_D = 0.0197. \quad (9)$$

The contours of the flowfield presented below are used to verify that the solution captures the expected transonic features of the ONERA M6 case at $M_{\infty} = 0.84$ and $\alpha = 3.06^\circ$.

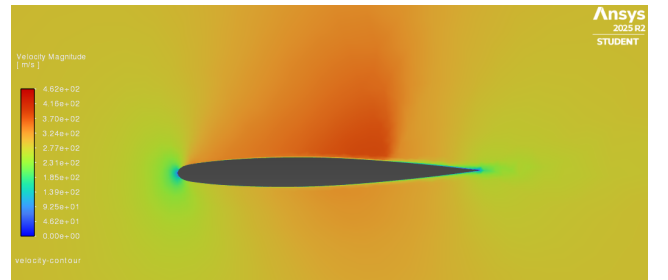


Fig. 16. Contour of magnitude of velocity on the symmetry plane. A high-speed flow develops on the upper surface while, due to the no-slip condition, near-wall velocities are close to zero. On the upper surface there is a sharp gradient in the velocity that indicates the presence of a transonic shock.

This is supported by the velocity contour, which indicates stagnation at the leading edge followed by a strong acceleration on the upper surface. The boundary layer near the wall is

manifest as a region of reduced velocity imposed by no-slip. Farther downstream on the upper surface, there is a narrow region of abrupt velocity change. Such behavior is consistent with shock formation in transonic compressible flow and is expected at $M_\infty = 0.84$.

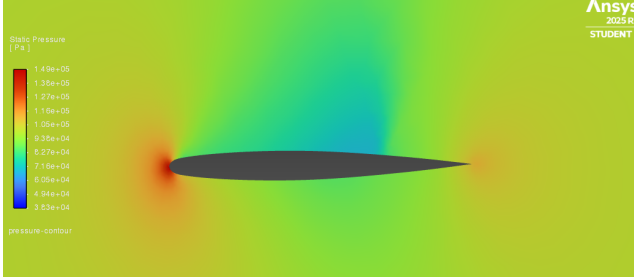


Fig. 17. Static pressure contour on the symmetry plane. High pressure occurs at the leading-edge stagnation region and relatively lower pressure develops over the upper surface, consistent with lift generation at a positive angle of attack.

The symmetry-plane pressure contour shows a high-pressure region at the leading edge and a lower-pressure region over the upper surface where the flow accelerates. The pressure difference between the lower and upper surfaces is the source of lift, generated by the upward pressure force on the lower surface and by suction on the upper surface. The pressure field is consistent with the velocity field and with the applied angle of attack.

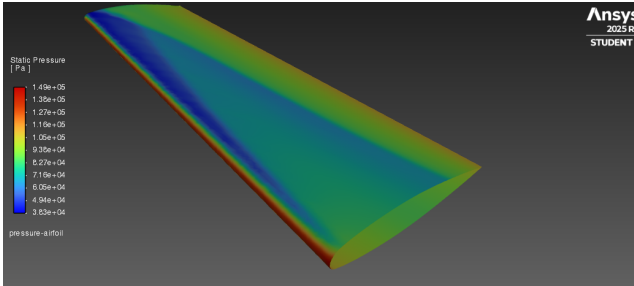


Fig. 18. Three-dimensional static pressure distribution on the wing surface. A strong pressure peak forms near the leading edge, followed by a pressure drop over the upper surface and a sharp pressure jump associated with shock formation.

The 3D surface pressure plot shows the spanwise structure of the transonic flow. Pressure is highest near the leading edge and then decreases along the upper surface. A distinct, narrow band of abrupt pressure change appears on the upper surface, corresponding to the shock. This shock structure is the primary transonic feature of the ONERA M6 case and is the main reason the case is helpful as a 3D validation benchmark.

h) Validation Against ONERA M6 Reference Data: To validate the integrated force coefficients, the computed C_L and C_D were compared to published ONERA M6 reference results across multiple solvers and mesh resolutions. The reference plots show solver-to-solver variation and the trends in lift and drag as mesh resolution increases.

The reference lift plot indicates that more refined simulations generally yield lift coefficients in the upper 0.26–0.27 range. The present simulation gives $C_L = 0.2539$, which is

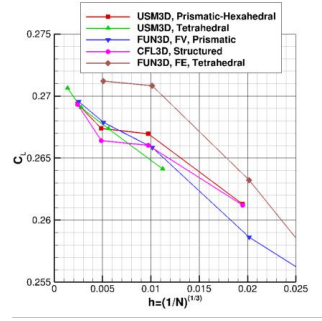


Fig. 19. Published ONERA M6 Case 2308 lift coefficient trends across multiple solvers and mesh resolutions. As mesh resolution increases, C_L trends toward approximately 0.27.

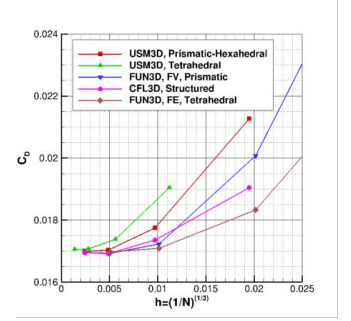


Fig. 20. Published ONERA M6 Case 2308 drag coefficient trends across multiple solvers and mesh resolutions. As the resolution increases, C_D approaches approximately 0.017.

slightly lower than the finest reference results. Given the mesh constraint of fewer than one million elements and the use of steady RANS with the Spalart–Allmaras model, a modest underprediction of lift relative to higher-resolution benchmarks is expected.

The reference drag plot indicates that drag typically trends downward with refinement, with higher-resolution solutions clustering around $C_D \approx 0.017$. The present simulation yields $C_D = 0.0197$, slightly higher than the most refined reference results. This deviation direction aligns with the typical behavior of coarser meshes and lower-order turbulence models, which tend to smear shocks and increase numerical dissipation, thereby increasing drag.

Overall, the computed lift and drag fall within the published solver spread and follow the same resolution-dependent trends seen in the reference data. This supports that the current setup is physically consistent and captures the correct transonic swept-wing behavior within the computational limits of the student solver.

i) Scope and Limitations: This analysis used steady Reynolds-averaged Navier–Stokes modeling and therefore does not capture unsteady shock motion or transient separation. Mesh resolution and near-wall treatment were selected to balance accuracy with computational constraints imposed by the student solver. Even with these limitations, the setup captures the dominant transonic swept-wing flow behavior and is sufficient for verifying the numerical approach used in this work.

IV. CONCLUSION

This paper provided a comparative review of conventional and non-conventional aircraft wing designs, emphasizing the influence of geometry, sweep, and flow regime on aerodynamic performance. Variable-sweep, forward-swept, delta, blended, and swept wings were evaluated using established aerodynamic theory as well as prior experimental and computational studies. The analysis indicated that no single wing configuration is optimal for all flight conditions. Rather, each design involves a trade-off among lift generation, drag, stability, structural complexity, and operational envelope.

The ONERA M6 computational case study reinforced these concepts by illustrating how swept-wing geometry controls transonic shock formation, lift distribution, and drag behavior at high subsonic Mach numbers. While the simulation was not used to validate performance metrics for the wing configurations discussed elsewhere in this paper, it serves as an example of how standardized benchmark cases are used in industry and research to assess numerical methods and interpret swept-wing flow physics under transonic conditions.

The findings demonstrate that advanced wing designs are determined by mission-specific requirements. Modern aerodynamic performance depends on balancing fluid mechanics, structural considerations, and control strategies, rather than maximizing a single performance metric.

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